# ANSWER KEY & MARKING SCHEME SET 1

**Full Syllabus (Chapters 1–13)** | Class 9 Science | Maximum Marks: 80

## SECTION — A (Objective Type) — Answers — 20 × 1 = 20

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 2 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 3 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 4 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 5 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 6 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 7 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 8 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 9 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 10 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 11 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 12 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 13 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 14 | **(d)** 3 x 10^5 | 2 x 10^30 / 6 x 10^24. |
| 15 | **(d)** 50 per cent | 30/60. |
| 16 | **(a)** 2 : 1 | 75/25 = 3; 60/40 = 1.5; ratio 2 : 1. |
| 17 | **(d)** $S^{2-}$ | Z = 16 is sulphur, 2 extra electrons. |
| 18 | **(b)** u / sqrt(2) | v squared = u squared - 2g(h/2) with h = u squared/2g. |
| 19 | **(a)** 1 : 4 | a is inversely proportional to m. |
| 20 | **(b)** 100 J | Half converted to PE. |

## SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | Area = 96 square cm (1); About 2 per cent (1). | 1+1 |
| 22 | Mass of solution = 240 g (1); 24/240 x 100 = 10 per cent (1). | 1+1 |
| 23 | 4 g H needs 32 g O (1); Water = 36 g; 8 g of oxygen left (1). | 1+1 |
| 24 | 63x + 65(100 - x) = 6350 (1); x = 75, so the heavier isotope is 25 per cent (1). | 1+1 |
| 25 | Harmonic mean 2 x 30 x 60/90 (1); = 40 km/h (1). | 1+1 |
| 26 | a = -2 m/s squared (1); t = 5 s, s = 25 m (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 6 × 3 = 18

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 27 | Mean = 42.6/3 (1); = 14.2 cm (1); Greatest deviation = 0.1 cm (1). | 1+1+1 |

| | | |
|---|---|---|
| **28** | 320 g contains 120 g salt and 200 g water (1); At 20 degrees Celsius 200 g water holds 50 g (1); Crystals = 120 - 50 = 70 g (1). | 1+1+1 |
| **29** | S: 1/32 = 0.03125 (1); O: 1.6/16 = 0.1 (1); Ratio about 1 : 3: $SO_3$ (1). | 1+1+1 |
| **30** | 2, 8, 3 (1); $M^{3+}$ (1); Mass = 54 + 48 = 102 u (1). | 1+1+1 |
| **31** | v = 20 m/s; s1 = 50 m (1); s2 = 200 m (1); t3 = 10 s; s3 = 100 m; total = 350 m (1). | 1+1+1 |
| | **OR:** a = 4 m/s squared (1); F = 6000 N, s = 50 m (1); Weight = 15000 N; force is 0.4 of weight (1). | 1+1+1 |
| **32** | PE lost = 200 J (1); KE gained = 90 J (1); Work against friction = 110 J (1). | 1+1+1 |
| | **OR:** Wavelengths 4 : 1 (1); Speeds equal (1); Loudness equal; pitch higher for 400 Hz (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers

**6 × 5 = 30**

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| **33** | Apparatus: thread, bob, stand, stopwatch, scale (1); Independent: length; dependent: time period (1); Constants: mass, small angle (1); Time 20 oscillations and repeat (1); Plot T against L and conclude that T increases with L (1). | 1+1+1+1+1 |
| **34** | NaCl 36 g stays dissolved at 20 degrees Celsius (solubility 36) (1); KNO3 remaining dissolved = 32 g (1); KNO3 crystallised = 110 - 32 = 78 g (1); Crystals are nearly pure KNO3 (1); Fractional crystallisation (1). | 1+1+1+1+1 |
| **35** | Run 1: O = 1.6 g, Mg : O = 3 : 2 (1); Run 2: O = 4.0 g, Mg : O = 3 : 2 (1); Same ratio in both runs (1); Formula MgO: 24 : 16 = 3 : 2 (1); Law and formula verified (1). | 1+1+1+1+1 |
| **36** | 69x + 71(100 - x) = 6980 (1); x = 60, so 60 per cent and 40 per cent (1); Neutrons: 38 and 40 (1); Element is gallium (1); Average mass between 69 and 71 (1). | 1+1+1+1+1 |
| | **OR:** Distances inversely proportional to a: 2 km and 1 km (1); v squared = 2 x 0.5 x 2000, v = 44.7 m/s (1); t1 = 89.4 s (1); t2 = 44.7 s; total 134 s (1); Check s2 = 44.7 squared/2 = 1000 m (1). | 1+1+1+1+1 |
| **37** | a = 3 m/s squared (1); v = 15 m/s (1); s = 37.5 m (1); p = 120 kg m/s (1); F = change in p/t = 24 N (1). | 1+1+1+1+1 |
| | **OR:** v = sqrt(2 x 10 x 3.2) = 8 m/s (1); KE at bottom = 64 J (1); Work by friction = -64 J (1); f x 4 = 64, f = 16 N (1); Energy conserved: PE to KE to heat (1). | 1+1+1+1+1 |
| **38** | lambda = 0.782 m (1); t = 344/344 = 1 s (1); Waves between source and wall = 172/0.782 = 220 (1); Echo path 344 m: 440 waves (1); Frequency equals the waves per second (1). | 1+1+1+1+1 |
| | **OR:** Cut equal potato cubes, weigh (1); Place in solutions for same time (1); Reweigh and calculate the percentage change (1); Plot the graph of percentage change vs concentration (1); Intercept on the x-axis gives isotonic concentration (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

## ANSWER KEY & MARKING SCHEME SET 2

**Full Syllabus (Chapters 1–13)** | Class 9 Science | Maximum Marks: 80

### SECTION — A (Objective Type) — Answers — 20 × 1 = 20

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** 0.68 m and 3 m | lambda = v/f. |
| 2 | **(b)** hypotonic to the cell | Water entered. |
| 3 | **(d)** accumulation of food | Phloem blocked. |
| 4 | **(c)** Fungi | Hyphae and chitin. |
| 5 | **(b)** 14 March | Mid cycle. |
| 6 | **(a)** 4000 J | 400 W x 10 s. |
| 7 | **(b)** the theory is strongly supported but remains open to tests | Science is provisional. |
| 8 | **(a)** only one component sublimes | Different behaviour on heating. |
| 9 | **(a)** 4.4 g | Mass of gas escaped. |
| 10 | **(d)** 24.3 u | (1896 + 250 + 286)/100 = 24.32. |
| 11 | **(d)** 8 km/h | 24 km/ 3 h. |
| 12 | **(a)** the pull of the book on the Earth | Third-law pair. |
| 13 | **(c)** 250 J | mgh. |
| 14 | **(d)** 24 m/s | 0.6 x 40. |
| 15 | **(d)** synthesis of ATP | Mitochondria do ATP. |
| 16 | **(b)** glandular epithelium | Secretion. |
| 17 | **(b)** They are in the same class and phylum | Higher ranks shared. |
| 18 | **(a)** the one with an eye grows | Bud needed. |
| 19 | **(a)** algal bloom and fish kill | Eutrophication. |
| 20 | **(d)** A is false but R is true. | A is false (theories can be refined); R is true. |

### SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | KE change = (1/2)(2)(25 - 9) (1); = 16 J (1). | 1+1 |
| 22 | Distance = 170 m (1); v = 170/0.5 = 340 m/s (1). | 1+1 |
| 23 | SA/V = 6 and 2 (1); Larger cell exchanges less efficiently, so it divides (1). | 1+1 |
| 24 | Secondary xylem accumulates every year (1); Lignified walls give strength (1). | 1+1 |
| 25 | New evidence from molecular studies (1); Discovery of new organisms (1). | 1+1 |
| 26 | Muscle 46, sperm 23 (1); Egg 23, zygote 46 (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 6 × 3 = 18

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 27 | 0.5 per cent: gains mass (1); 5 per cent: loses mass (1); 1 per cent is isotonic with the potato cell sap (1). | 1+1+1 |
| 28 | Potato turns blue-black as it stores starch (1); Carrot gives less blue-black (1); Parenchyma stores food (1). | 1+1+1 |
| 29 | Feathers? yes: pigeon (1); No: has hair? yes: bat (1); No: gills with scales: shark; moist skin: frog; dry scaly skin: snake (1). | 1+1+1 |
| 30 | Ovulation about 14 days before the next period: day 16 (1); Fertile period: days 13 to 17 (1); Sperms survive 3 days before (1). | 1+1+1 |
| 31 | Annual emission = 250 x 12 x 0.8 = 2400 kg (1); Trees = 2400/22 (1); = 109 trees (1). | 1+1+1 |
| | **OR:** Shows how reliable the result is (1); e.g. 12.3 +/- 0.1 cm (1); Allows comparison with other results (1). | 1+1+1 |
| 32 | A: 5/55 = 9.1 per cent (1); B: 8/108 = 7.4 per cent (1); A is more concentrated (1). | 1+1+1 |

| | | |
|---|---|---|
| | **OR:** Mg2+ and O2- attract more strongly (1); More energy to break the lattice (1); Hence higher melting point (about 2800 vs 801 degrees Celsius) (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers

**6 × 5 = 30**

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| **33** | Place a twig in red ink (1); Section shows red xylem (1); Girdle a branch (1); Bulge above the ring (1); Conclusions about xylem and phloem (1). | 1+1+1+1+1 |
| **34** | Cell type: prokaryote - bacterium (1); Eukaryote unicellular: Amoeba (no wall), yeast (wall) (1); Multicellular plants: vascular? no: moss; yes: seeds? no: fern (1); Seeds: cones - pine; flowers - rose (dicot), wheat (monocot) (1); Characters: cell wall, vascular tissue, seeds, cotyledon (1). | 1+1+1+1+1 |
| **35** | Tube 1: moist cotton, room temperature (1); Tube 2: dry cotton (1); Tube 3: seeds submerged in boiled water with oil (1); Tube 4: moist cotton in a fridge (1); Only tube 1 germinates; conclusion (1). | 1+1+1+1+1 |
| **36** | Absorbed = 0.7 x 342 = 239 W/m squared (1); At equilibrium the outgoing is 239 W/m squared (1); Albedo 0.28: absorbed 246 W/m squared, warming (1); More greenhouse gases reduce the outgoing radiation (1); Surface warms until balance is restored (1). | 1+1+1+1+1 |
| | **OR:** Same mass and volume (1); Vary only temperature (1); Record times (1); Graph of time vs temperature (1); Errors: reaction time and stirring (1). | 1+1+1+1+1 |
| **37** | Particles carry charge and repel (1); Coagulation with electrolytes (1); Electrophoresis moves particles in a field (1); Dialysis separates dissolved ions (1); Examples: blood purification (1). | 1+1+1+1+1 |
| | **OR:** Atoms of different elements have different masses (1); 12 g of Mg has half as many atoms as 12 g of C (1); Number of atoms is proportional to mass divided by atomic mass (1); 24 g of Mg has as many atoms as 12 g of C (1); Atomic masses link mass to number of atoms (1). | 1+1+1+1+1 |
| **38** | n = 24 - 12 = 12 (1); B: p = 11, n = 12, A = 23 (1); B is sodium $^{23}_{11}\mathrm{Na}$ (1); 2, 8, 1 (1); A is magnesium-24 (1). | 1+1+1+1+1 |
| | **OR:** T = 2 x 3.14/4 = 1.57 s (1); Speed constant 4 m/s (1); Velocity changes direction (1); Displacement in half period 2 m (1); Average velocity = 2/0.785 = 2.55 m/s (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER

# ANSWER KEY & MARKING SCHEME SET 3

**Full Syllabus (Chapters 1–13)** | Class 9 Science | Maximum Marks: 80

## SECTION — A (Objective Type) — Answers — 20 × 1 = 20

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 2 | **(c)** A is true but R is false. | A is true; R is false because the gas escapes. |
| 3 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 4 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 5 | **(b)** 4 m/s squared | F/m. |
| 6 | **(a)** 480 W | 2400/5. |
| 7 | **(d)** 150 m | 1500 x 0.2/2. |
| 8 | **(c)** cell wall | Rigid wall. |
| 9 | **(d)** sensory neuron, relay neuron and motor neuron | Reflex arc. |
| 10 | **(a)** animal | Animalia. |
| 11 | **(c)** wings or hairs | Light structures. |
| 12 | **(a)** the ground is heated by sunlight and warms the air | Surface heating. |
| 13 | **(d)** 5.4 cm | Subtract the zero error. |
| 14 | **(a)** 32 g | Water 100 g holds 32 g; crystals 64 - 32. |
| 15 | **(c)** 3 : 8, $CO_2$ | Atoms ratio 2.4/12 : 6.4/16 = 1 : 2. |
| 16 | **(b)** 20 and 16 | X lost 2, Y gained 2. |
| 17 | **(a)** 1.5 t | v = at; retard time = at/2a = t/2. |
| 18 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 19 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 20 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |

## SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | Absorbed = 0.7 x 342 (1); = 239 W/m squared (1). | 1+1 |
| 22 | A straight line shows proportionality (1); Easy to read the slope and spot errors (1). | 1+1 |
| 23 | Boiling point is fixed (1); Heat supplies latent heat (1). | 1+1 |
| 24 | Na+: 2, 8; F-: 2, 8 (1); Same as neon (1). | 1+1 |
| 25 | Maximum 2n squared with n = 1 is 2 (1); K shell is the smallest and nearest to the nucleus (1). | 1+1 |
| 26 | Dropped: 2 s (1); Thrown down: 20 = 10t + 5t squared gives t = 1.24 s (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 6 × 3 = 18

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 27 | p + n = 31 and n = p + 5 (1); 2p = 26, so p = 13 and n = 18 (1); Z = 13; configuration 2, 8, 3 (1). | 1+1+1 |
| 28 | s1 = a/2 = 2, so a = 4 m/s squared (1); s4 = 0.5 x 4 x 16 (1); = 32 m (1). | 1+1+1 |
| 29 | a = v squared/2s = 2 x 10^6 m/s squared (1); F = 0.01 x 2 x 10^6 = 20000 N (1); t = v/a = 0.0002 s (1). | 1+1+1 |
| 30 | Energy proportional to x squared (1); Doubling the extension gives 4 times the energy (1); 16 J (1). | 1+1+1 |

| | | |
|---|---|---|
| **31** | T = 0.00143 s (1); lambda = 0.486 m (1); Waves in 5 s = 3500 (1). | 1+1+1 |
| | **OR:** Break cells gently (1); Centrifuge at low speed to remove nuclei (1); Higher speed to get mitochondria (1). | 1+1+1 |
| **32** | Constant energy demand (1); Continuous contraction (1); Supplies ATP (1). | 1+1+1 |
| | **OR:** Moss: gametophyte dominant (1); Fern: sporophyte dominant (1); Both need water for fertilisation (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers

**6 × 5 = 30**

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| **33** | Change = 0.16 x 70 = 11.2 kg m/s (1); Impulse = 11.2 N s (1); F = 560 N (1); Doubling time halves force (1); Same impulse (1). | 1+1+1+1+1 |
| **34** | 3000 kW x 24 = 72000 kWh (1); Cost = 3.6 lakh rupees (1); Number of bulbs = 3 x 10^6/100 (1); = 30000 bulbs (1); Energy = P x t (1). | 1+1+1+1+1 |
| **35** | Time gap less than 0.1 s in a small room (1); A big hall delays reflections (1); Use of absorbers (1); Curved ceilings and sound boards (1); Ceiling height above 17 m avoided or treated (1). | 1+1+1+1+1 |
| **36** | Stem cells unspecialised (1); Differentiation into RBC, nerve, muscle (1); Plant meristems (1); Uses: bone marrow transplants (1); Ethical issues: source of embryonic cells (1). | 1+1+1+1+1 |
| | **OR:** Xerophytes: thick cuticle, sunken stomata (1); Hydrophytes: aerenchyma (1); Mesophytes: normal tissues (1); Root tissues for water (1); Support tissues (1). | 1+1+1+1+1 |
| **37** | Roots to absorb water (1); Vascular tissue for transport (1); Seeds protect the embryo (1); Flowers for pollination (1); Fruits for dispersal (1). | 1+1+1+1+1 |
| | **OR:** Causes: blocked tubes, low sperm count (1); Hormonal problems (1); IVF steps (1); Artificial insemination (1); Social acceptance and ethics (1). | 1+1+1+1+1 |
| **38** | Less transpiration, less rainfall (1); Carbon released (1); Albedo changes (1); Biodiversity loss (1); Soil erosion and floods (1). | 1+1+1+1+1 |
| | **OR:** Strength: evidence-based (1); Strength: self-correcting (1); Limitation: cannot test untestable ideas (1); Chance: penicillin discovery (1); Persistence: many experiments repeated (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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